Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/28/2005
PI Name:	Grugel, Richard N. Ph.D.		
Project Title:	Toward Understanding Pore Formation and Mobility during Controlled Directional Solidification (PFMI)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Marshall Space Flight Center		
PI Address 1:	Mail Code SD46		
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PI Web Page:	http://www1.msfc.nasa.gov/		
City:	MSFC	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2001	**End Date:**	02/02/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Volz, Martin	**Contact Phone:**	256 544-5078
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Anilkumar, A. V. (Vanderbilt University)		
Grant/Contract No.:	Intramural		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A number of SCN and SCN-water alloys have been directionally melted and re-solidified in the PFMI apparatus within the International Space Station (ISS) Glovebox facility. During processing the solid/liquid interface is monitored and recorded to observe bubble and other solidification interactions and phenomena. The Pore Formation and Mobility Investigation melts samples of a transparent modeling material, succinonitrile (SCN) and succinonitrile water mixtures in a controlled, directional manner. Observations of how bubbles form in the samples, how they move, and how they interact will be studied with the intent of understanding how to minimize their detrimental effects. On Earth when scientists melt metals, bubbles that form in the molten material can rise to the surface, pop, and		

	disappear. In the microgravity environment aboard the International Space Station bubbles do not rise and disappear. Prior space experiments have shown that bubbles often become trapped in the final metal or crystal sample. In the solid, these bubbles, or porosity, are defects that diminish both the material's strength and usefulness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work to date has shed light on conducting and interpreting ground-based experimental results pertinent to understanding and modeling solidification phenomena.
Task Progress:	Experimental runs of the PFMI samples have, to best of our knowledge, been completed. Seventeen experiments were performed using a combination of samples containing pure succinonitrile with nitrogen and samples with succinonitrile, nitrogen, and small (up to 1 wt. %) amounts of water. The transparent nature of the samples allowed for direct observation and recording of processing events during the controlled melt back and re-solidification of the samples. Controlled melt back of the established planar solid/liquid interface has revealed a number of microgravity dependent phenomena regarding bubble movements and interactions. These include quantitative evaluation of bubbles velocities moving up an imposed temperature gradients and numerous other observations of thermocapillary effects otherwise masked by Earth's gravitational field. Controlled re-solidification of the melted sample provided information, and the first theoretical agreement, regarding plane front dynamics and the observation of the little studied Liquid to Solid + Gas invariant transformation. The fifteen sample ampoules and 56 video tapes are still aboard ISS after which their return will provide additional data for evaluation. The European Space Agency has also expressed interest in utilizing/adapting the PFMI apparatus for their use. Several proceedings papers and presentations have been made.
Bibliography Type:	Description: (Last Updated: 10/11/2016)
Books/Book Chapters	Grugel RN, Anilkumar AV, Lee CP. "Direct observation of pore formation and bubble mobility during controlled melting and re-solidification in microgravity." in "Solidification Processes and Microstructures: A Symposium in Honor of Wilfried Kurz. from 2004 TMS Annual Meeting, Charlotte, North Carolina, March 14-18, 2004." Ed. M. Rappaz, C. Beckermann, R. Trivedi. Warrendale, PA: The Metallurgical Society, 2004, p. 111-116. http://www.wiley.com/WileyCDA/WileyTitle/productCd-0873395727.html , Mar-2004
Papers from Meeting Proceedings	Grugel RN, Anilkumar AV, Cox MC. "Observation of an Aligned Gas - Solid 'Eutectic' During Controlled Directional Solidification Aboard the International Space Station - Comparison with Ground-based Studies." 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 10-13, 2005. 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 10-13, 2005. AIAA paper 2005-919. http://dx.doi.org/10.2514/6.2005-919 , Jan-2005
Papers from Meeting Proceedings	Strutzenberg LL, Grugel RN, Trivedi R. "Morphological Evolution of Directional Solidification Interfaces in Microgravity: An Analysis of Model Experiments Performed on the International Space Station." 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 10-13, 2005. 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 10-13, 2005. AIAA paper 2005-917. http://dx.doi.org/10.2514/6.2005-917 , Jan-2005
Presentation	R.N. Grugel, A.V. Anilkumar and C.P. Lee "Direct Observation of Pore Formation and bubble mobility during controlled melting and re-solidification in microgravity" Metallurgical Society, Charlotte, NC, 15 March 2004 Mar-2004
Presentation	Grugel, R.N., P. Luz, G.A. Smith, R. Spivey, C. Mingo, S. Sen, L. Jeter, M. Volz, and A.V. Anilkumar. "The Pore Formation and Mobility Investigation: The Apparatus, Operations, Science Obtained (to Date), and Potential for Continued Usage" European Space Agency Headquarters, Paris, France, Invited 5-6 April Apr-2004
Presentation	Grugel, R.N. "Materials Processing in Space: Model Experiments aboard the International Space Station" Mechanical Engineering Department of Vanderbilt University, Nashville, TN, 19 April 2004, Invited. Apr-2004
Presentation	Grugel, R.N., A. V. Anilkumar and M. C. Cox. "Observation of an Aligned Gas - Solid 'Eutectic' During Controlled Directional Solidification Aboard the International Space Station - Comparison with Ground-based Studies" 43rd AIAA Aerospace Sciences Meeting and Exhibit, 10-13 January 2005, Reno, Nevada Jan-2005
Presentation	Grugel, R.N., L.L. Strutzenberg and R. Trivedi "Morphological Evolution of Directional Solidification Interfaces in Microgravity: An Analysis of Model Experiments Performed on the International Space Station" 43rd AIAA Aerospace Sciences Meeting and Exhibit, 10-13 January 2005, Reno, Nevada Jan-2005